DAILY POWER GENERATION ANALYSIS

1. Project Overview and Objective

This project analyzes daily power generation data across Indian states for the period 2019–2020. The study focuses on comparing planned generation with actual generation to evaluate plant performance and operational efficiency.

The data includes both categorical variables such as region, state, sector, power station type, and power station unit, as well as numerical variables such as installed capacity, planned generation, and actual daily power generation.

The project involves data cleaning and pre processing using Excel, followed by data analysis and visualization using Power BI. The aim is to understand power generation trends, regional performance, and the gap between planned and actual generation. Key performance indicators such as generation variance, performance percentage, and capacity utilization were calculated to measure deviations and efficiency levels. Excel was used for data cleaning, statistical analysis, KPI calculation, and dashboard creation

- To analyze daily power generation trends for the years 2019–2020.
- To compare planned (program) generation with actual generation.
- To calculate generation variance to measure deviations from target
- To compare planned power generation versus actual generation across power stations.
- To evaluate state-wise and region-wise power generation performance.
- To evaluate plant performance using performance percentage.
- To assess operational efficiency using capacity utilization.
- To identify overperforming and underperforming power plants.
- To create a dashboard summarizing key performance indicators for decision-making.

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2. Data Sources

- **Source Description and Timeline:** India Data Portal / Source -Ministry of Power/ Energy - https://indiadataportal.com/p/power/r/mop-power_generation-pl-dl-abc
- **Domain**: Energy / Renewable Energy (India)

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2. Problem Statement

The daily power generation data set contains a large volume of structured data related to electricity generation across different states, regions, and power stations in India. Although the data is well-organized, it includes variations in daily generation values, zero generation records, and differences between planned and actual power generation, which make direct interpretation difficult. Without proper data cleaning and analysis, it is challenging to

understand power generation trends, regional performance differences, and capacity utilization of power stations.

Therefore, there is a need to systematically clean, process, and analyze this structured data set using analytical tools such as Excel and Power BI in order to extract meaningful insights, identify performance gaps, and support data-driven understanding of daily power generation in India.

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4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
ID	Integer	Unique identifier for each record in the dataset
Date	Date	The date of power generation measurement
Region	Text(String)	The geographical region where the power station is located (e.g., Northern).
State_name	Text(String)	The name of the state in India where the station is located.
State_code	String/Integer	A code representing the state.
Sector	Text(String)	Type of ownership of the power station.
Station_type	Text(String)	Type of power station based on energy source.
Power_station	Text(String)	The name of the specific power station.
Power_station_unit	Text(String)	The specific unit or generator within the power station.
Monitored_capacity	Float(Decimal No)	The total installed or monitored capacity of the unit in MW.
Todays_gen_prgm	Float(Decimal No)	Planned generation for that dayin MW.
Todays_gen_act	Float(Decimal No)	Actual generation achieved for that day in MW.
Gen_Variance	Float(Decimal No)	Variance b/w actual and planned
Performance	Percentage	Format as percentage
Capacity MU perday	Decimal Number	Numerical value represent capacity unit per day
Capacity Utilization MU	Decimal Number	Represent Utilization percentage
Performance Status	Text/String	Contains Category value like over performed, under performed

Utilization Category	Text/String	Contains category like low, moderate, high

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5. Tools & Technologies

- **Excel:** Data cleaning, transformation.
- **Power BI:** Dimension Table, Data modelling, DAX calculations, visualization, and interactive dashboard creation.

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6. Data Pre-Processing (Excel / Power Query)

Data Cleaning & Transformation:

1. Duplicates Check:

Duplicate records were identified and removed based on Date, State Name, Station Type and power station to ensure each daily station record was unique.

2. Handle Missing Value:

Date Column:

- Applied TRIM () function to remove leading and trailing spaces and CLEAN () function to remove non-printable and unwanted characters.
- Converted the column to proper Date format DD-MM-YYY using Format Cells.
- Verified that the dataset contained complete records from 2019 to October 2020.

State_name Column:

- The State_Name column was standardized by applying the TRIM() and CLEAN() functions to remove extra spaces and non-printable characters.
- Some states has spelling error, so used find and replace .

Sector:

- Those records contained non-informative categorical values such as 'None' in analytical fields like sector and station type, I removed them to maintain data consistency and improve visualization clarity.

Power Station:

- Applied TRIM() and CLEAN() function to remove non- printable and unwanted characters and then applied = proper function to standardize text.

Numeric Column:

- Used Filter option to identify blank or zero values. Zero values in numeric columns were retained as they represent valid operational scenarios such as plant shutdowns or no planned generation, and therefore were not treated as missing data. No artificial values were inserted.

  0 in Monitored Capacity - plants not operational,

0 in Today Gen Program-no planned generation,

0 in Today Gen Actual - no electricity generated.

3. **Data Types:**

- Converted Date column to Date format.
- Converted generation and capacity columns to Whole Number.
- Converted performance column to Percentage format.
- Ensured text columns such as State and Power Station Name were properly formatted.

4. Calculated Columns

General Variance : Today Generation Actual - Today Generation Program

difference between planned generation and actual generation.Its helps to identify whether a power plant is performing above or below. If variance is + produced more than planned , if variance is negative produced less than planned. Negative variance helps identify power shortage days, poor performing station, operational inefficiency.

Performance: (today generation actual/today generation program)

This shows how efficiently the plant achieved its planned target. It shows comparision between different plant regardless size.

Capacity MU Per day: Monitored capacity*24/1000

Capacity MU per day was calculated by converting monitored capacity from MW to MU using the formula Capacity. This represents the maximum possible energy a plant can generate in one day.

Capacity Utilization MU: (Today Actual/Capacity MU Per Day)

Compares actual vs. maximum. Capacity Utilization measures how much of the plants maximum possible capacity is actually used

Performance Status: Performance Status =IF(Performance=0,"No Generation",IF(Performance>=1,"Over Performed",IF(Performance>=0.95,"On Target","Under Performed")))

Based on the performance categorizes plants into Over performed, On Target, or Under performed. This makes analysis easier for management.

Utilization Category: =IF(CU>=0.85,"High",IF(CU>=0.5,"Moderate","Low"))

Based on Capacity Utilization Utilization Category groups plants into High, Moderate, or Low utilization levels. It helps identify which plants are efficiently operating and which require attention.

5. Descriptive Statistics

Today_Gen_Program	
Mean	2.738313455
Standard Error	0.006860759
Median	1.14
Mode	0
Standard Deviation	3.724507591
Sample Variance	13.8719568
Kurtosis	7.2322847
Skewness	2.2264619
Range	34.67
Minimum	0
Maximum	34.67
Sum	807005.62
Count	294709

Today_Gen_Actual	
Mean	2.672246419
Standard Error	0.007807612
Median	0.69
Mode	0
Standard Deviation	4.238526672
Sample Variance	17.96510835
Kurtosis	14.30831497
Skewness	2.666666285
Range	150
Minimum	0
Maximum	150
Sum	787535.07
Count	294709

For descriptive Statistics selected Today Generation Program and Actual columns because they represent planned and real production values,using Analytics Tool pak on the Daily Power Generation. Descriptive statistics were used to summarize data using measures such as mean, minimum, maximum, and standard deviation.

6. Conditional Formatting: Conditional formatting applied to Performance, Capacity Utilization and General variance, Highlight cell rules were applied to Generation Variance & performance status

7. Table Structure Validation

- Ensured first row contained proper column headers.
- Converted dataset into Excel Table (Ctrl + T) to allow correct import into power BI.

DIMENSION TABLE USING POWER BI

The data set was first cleaned and transformed in Microsoft Excel by removing duplicates, handling missing values, standardizing formats, and creating calculated columns. After cleaning, the Excel file was imported into Power BI using the 'Get Data from Excel' option. The main data set was loaded as the Fact_Power Generation table

DIM DATE:

Created a Dim Date table using the CALENDAR function to generate a continuous list of dates between the minimum and maximum dates in the data set. ADDCOLUMNS was used to extract Year, Month, Month Year

This table enables time-based analysis such as daily, monthly, and yearly power generation trends and allows proper relationship with the fact table.

Date	Year	Month Number	Month Name	Month Year
01 January 2019	2019	1	January	Jan 2019
02 January 2019	2019	1	January	Jan 2019
03 January 2019	2019	1	January	Jan 2019
04 January 2019	2019	1	January	Jan 2019
05 January 2019	2019	1	January	Jan 2019
06 January 2019	2019	1	January	Jan 2019
07 January 2019	2019	1	January	Jan 2019
08 January 2019	2019	1	January	Jan 2019
09 January 2019	2019	1	January	Jan 2019
10 January 2019	2019	1	January	Jan 2019
11 January 2019	2019	1	January	Jan 2019
12 January 2019	2019	1	January	Jan 2019
13 January 2019	2019	1	January	Jan 2019
14 January 2019	2019	1	January	Jan 2019
15 January 2019	2019	1	January	Jan 2019
16 January 2019	2019	1	January	Jan 2019
17 January 2019	2019	1	January	Jan 2019
18 January 2019	2019	1	January	Jan 2019
19 January 2019	2019	1	January	Jan 2019
20 January 2019	2019	1	January	Jan 2019
21 January 2019	2019	1	January	Jan 2019
22 January 2019	2019	1	January	Jan 2019
23 January 2019	2019	1	January	Jan 2019
24 January 2019	2019	1	January	Jan 2019

Columns Created

- Date
- Year
- Month Number
- Month Name
- Month Year

DAX Formula used

```
Dim_Date
=ADDCOLUMNS(CALENDAR(min(Fact_PowerGeneration[date]),MAX(Fact_PowerGeneration[date])),"Year",YEAR([Date]),"Month Number",MONTH([Date]),"Month Name",FORMAT([Date],"MMMM"),"Month Year",FORMAT([Date],"MMM YYYY"))
```

DIM_STATE

The DimState table contains unique state names to analyze state-wise power generation performance. Created the Dim State table by referencing the fact table and removing duplicate state values to ensure each state appears only once.

The main purpose of creating the Dim_State table is to enable state-wise analysis of power generation, which helps in comparing generation performance between different states and identifying high-performing and low-performing states.

Columns included in state:

- Contains the unique name of each state.
- Duplicate state names were removed from the original data set to ensure each state appears only once.
- Represents the geographical region such as North, South, East, or West. Helps in performing regional analysis.

state_name	region
Delhi	Northern
Haryana	Northern
Himachal Pradesh	Northern
Jammu And Kashmir	Northern
Punjab	Northern
Rajasthan	Northern
Uttarakhand	Northern
Uttar Pradesh	Northern
Chhatisgarh	Western
Goa	Western
Gujarat	Western
Madhya Pradesh	Western
Maharashtra	Western
Andhra Pradesh	Southern
Karnataka	Southern
Kerala	Southern
Puducherry	Southern
Tamil Nadu	Southern
Telangana	Southern
Andaman And Nicobar Islands	Eastern
Bihar	Eastern
Jharkhand	Eastern
Odisha	Eastern
Sikkim	Eastern

DIM_POWER STATION

The Dim_Power Station is a dimension table that contains unique information about each power station in the data set.The purpose of this table is to enable power station-wise analysis, allowing comparison of generation performance between different power stations.

power_station	state_name	station_type	monitored_capacity(MW)
Rajghat Tps	Delhi	Thermal	67.5
I.P.Ccpp	Delhi	Ther (Gt)	270
Pragati Ccgt-Iii	Delhi	Ther (Gt)	1500
Pragati Ccpp	Delhi	Ther (Gt)	330.4
Rithala Ccpp	Delhi	Ther (Gt)	108
Badarpur Tps	Delhi	Thermal	0
Panipat Tps	Haryana	Thermal	210
Rajiv Gandhi Tps	Haryana	Thermal	600
Yamuna Nagar Tps	Haryana	Thermal	300
Mahatma Gandhi Tps	Haryana	Thermal	660
Indira Gandhi Stpp	Haryana	Thermal	500
Faridabad Ccpp	Haryana	Ther (Gt)	431.59
Bassi Hps	Himachal Pradesh	Hydro	66
Bhakra Left Hps	Himachal Pradesh	Hydro	594
Bhakra Right Hps	Himachal Pradesh	Hydro	785
Dehar H P S	Himachal Pradesh	Hydro	990
Giri Bata Hps	Himachal Pradesh	Hydro	60
Kashang Integrated Hep	Himachal Pradesh	Hydro	195
Larji Hps	Himachal Pradesh	Hydro	126
Pong H P S	Himachal Pradesh	Hydro	396
Sainj Hps	Himachal Pradesh	Hydro	100
Sanjay Hps	Himachal Pradesh	Hydro	120
Sawra Kuddu Hps	Himachal Pradesh	Hydro	0
Shanan Hps	Himachal Pradesh	Hydro	110

Columns included in power station

Power station name:

- Contains the unique name of each power station.
- Duplicate values were removed.

State :

- Identifies the state where the power station is located.
- Helps in linking stations to their respective states.

Station Type:

- Thermal, Hydro, Nuclear, Solar, Wind
- This helps analyze generation based on station type.

Installed Capacity

- Used to calculate capacity utilization.

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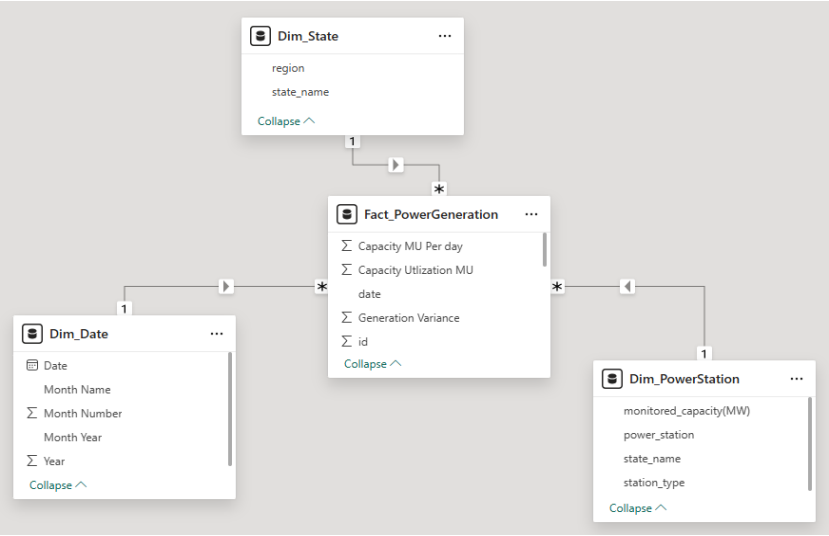
8. DATA MODELLING

DATA MODEL VIEW AND RELATIONSHIP

- A star schema data model was created by connecting the Fact Table with Dim_Date, Dim_State, and Dim_Power Station using one-to-many relationships.
- The Fact Table contains daily power generation values, while the dimension tables store descriptive information such as date, state, and power station.
- This data model improves data organization, enables efficient filtering, and supports accurate analysis and visualization in Power BI dashboards.
- Below shows the relationship diagram as fact table in the centre connecting to all dimension table.



DAX Measures:

DAX (Data Analysis Expressions) is used in Power BI to create measures and calculated for data analysis.

In this project, DAX measures were created to analyze power generation performance, compare planned vs actual generation and evaluate operational efficiency.

- **Total Actual Generation**

Purpose: Calculates the total electricity actually generated.

DAX: Total Actual Generation = sum(Fact_PowerGeneration[todays_gen_act(MU)])

- **Total Planned Generation**

Purpose: Calculates the total planned/program generation.

DAX: Total Planned Generation = sum(Fact_PowerGeneration[todays_gen_prgm(MU)])

- **Generation Variance**

Purpose: Calculates the difference between actual and planned generation.

DAX: General Variance = [Total Actual Generation]-[Total Planned Generation]

- **Total Monitored Capacity**

Purpose: Calculates total installed/monitored capacity.

DAX: Total Monitored Capacity = sum(Fact_PowerGeneration[monitored_capacity(MW)])

Performance %

Purpose: measures the average efficiency of power generation by comparing the actual generation with the planned generation across all power stations and dates..

DAX: Performance = DIVIDE([Total Actual Generation],[Total Planned Generation],0)*100

- **Capacity Utilization**

- Purpose: Measures how efficiently the installed capacity is used.

DAX: Capacity Utilization = DIVIDE([Total Actual Generation],[Total Monitored Capacity])

- **Average Generation Per Day**

Purpose: Calculates average daily power generation.

DAX: Average Generation Per Day = AVERAGE(Fact_PowerGeneration[todays_gen_act(MU)])

- **Total Power Stations**

Purpose: Counts the number of unique power stations.

DAX: Total Power Station = DISTINCTCOUNT(Fact_PowerGeneration[power_station])

- **Year-To-Date (YTD) Generation**

Purpose: cumulative performance over time and compare overall generation progress within the year.

DAX: Year-To-Date(YTD) Generation = TOTALYTD([Total Actual Generation],Dim_Date[Date])

❖ **Cumulative Generation**

Purpose: Cumulative generation shows the running total of daily actual power generation over time, starting from the first date to the selected date.Its helps to analyze how total power production increased throughout the year.

DAX: Cumulative Generation = CALCULATE([Total Actual Generation],FILTER(ALL(Dim_Date),Dim_Date[Date]<=MAX(Dim_Date[Date])))

❖ **Power Station Contribution %**

Purpose: Its shows the percentage share of total power generation produced by each power station.

DAX: Contribution PW % = DIVIDE([Total Actual Generation],CALCULATE([Total Actual Generation],ALL(Dim_PowerStation[power_station])),0)*100

❖ **State Contribution %**

Purpose : percentage share of total power generation contributed by each state.

DAX: Contribution State % = DIVIDE([Total Actual Generation],CALCULATE([Total Actual Generation],ALL(Dim_PowerStation[state_name])),0)*100

❖ **Station Type Contribution %**

Purpose : percentage share of total power generation contributed by each station.

DAX: Contribution Power Station Type = DIVIDE([Total Actual Generation (MU)],CALCULATE([Total Actual Generation (MU)],ALL(Dim_PowerStation[station_type])),0)*100

Gauge

Purpose based on the performance

DAX Gauge Maximum = 120, Gauge Minimum = 0, Target = 100

Performance Category :

Purpose To identify which station is underperformed.

DAX Performance Category =

```
VAR Actual = [Total Actual Generation (MU)]
VAR Planned = [Total Planned Generation (MU)]
RETURN
SWITCH(
   TRUE(),
   Actual = 0, "No Generation",
```

```
    Actual = Planned, "On Target",
    Actual > Planned, "Overperformed",
    Actual < Planned, "Underperformed",
    BLANK()
)
```

Station Rank :

Purpose :identify top and bottom performing station

DAX Station Rank = RANKX(ALL(Dim_PowerStation[power_station]),[Total Actual Generation (MU)],,DESC)

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8. Analysis and Visualizations (Power BI)

KPI Overiew

A total of 5 KPI includes using card visuals

a) Total Actual Generation (MU) – 787.54K

This KPI shows the total electricity actually generated during the selected time period (2019–2020).
It represents the real production output measured in Million Units (MU).

- To measure overall production performance.
- To compare with planned generation002E

b) Total Planned Generation (MU) – 807.01K

This KPI shows the target electricity generation planned by authorities.:

- Acts as a benchmark.
- Used to measure whether production met expectations.

c) Performance – 97.59%

This represents efficiency of actual generation compared to planned generation.

The system achieved 97.59% of its planned target, meaning it performed slightly below full target but still at a very high efficiency level.

d) Total Generation Variance – (-19.47K)

This shows the difference between planned and actual generation.
A negative variance means actual generation was lower than planned by 19.47K MU.

e) Capacity Utilization – 0.01

This represents how effectively installed capacity was utilized.

- Measures operational efficiency.
- Indicates whether resources are underutilized.

f) Total Power Stations – 514

Total number of power stations included in analysis.

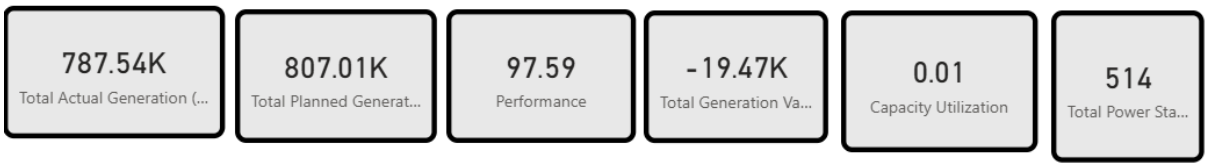
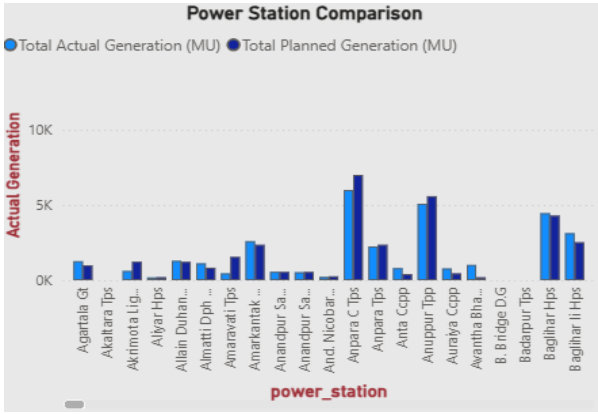


CHART ANALYSIS

Power Station Comparison : Clustered Column Chart



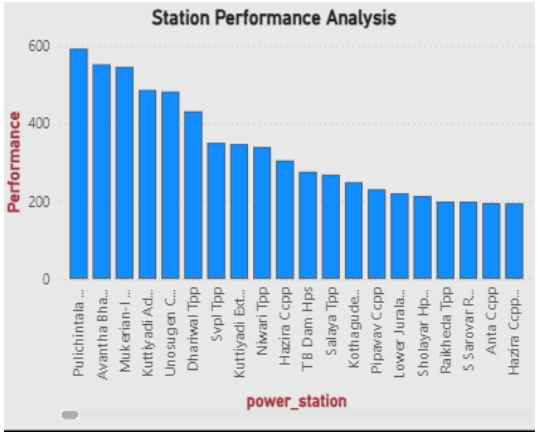
From the chart, it is evident that power generation is not evenly distributed across all stations. A small number of stations contribute significantly higher generation compared to others. Some stations consistently show actual generation close to planned values, indicating strong operational stability and efficient management. However, certain stations show noticeable gaps between planned and actual generation, indicating possible:

- Maintenance issues
- Fuel supply constraints
- Operational inefficiencies
- Seasonal dependency (hydro stations)

Insights:

✓ A few power stations contribute major share of total generation.
✓ Most stations operate near planned targets, reflecting good overall planning.
✓ Stations with high variance need operational review.
✓ Performance imbalance indicates opportunity for optimization.

Station performance analysis : Bar Chart



This chart evaluates efficiency of each power station using performance%

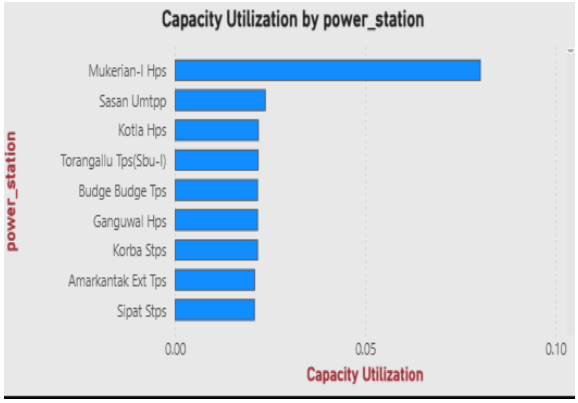
Performance percentage close to 100% indicates strong operational efficiency.

- Stations above 100% indicate over achievement.
- Stations below 95% may require operational attention.
- The majority of stations fall within the 95–100% range, suggesting stable performance.

Insights

✓ Overall station performance is high and consistent.
✓ Few stations underperform slightly, but not significantly.
✓ No major operational failure observed.
✓ Strong alignment between planning and execution.

Capacity Utlization By power station - Bar Chart



Stations with high capacity utilization are operating near their full potential.

Stations with low capacity utilization may indicate:

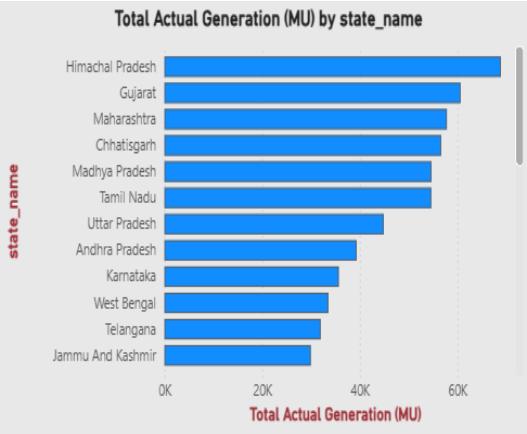
- Demand fluctuation
- Fuel shortage
- Scheduled shutdown
- Technical inefficiency

In daily power generation analysis, some stations show consistent utilization, while others fluctuate depending on season (especially hydro stations).

Insights

✓ Majority of stations operate efficiently.
✓ Some stations show under-utilization, indicating scope for improvement.
✓ Seasonal variation may impact utilization rates.
✓ Efficient capacity management is critical for energy planning.

Total Actual Generation by state - Column chart



This chart shows contribution of each state to total power generation. Some states such as Himachal Pradesh, Gujarat, Maharashtra, and others contribute significantly higher electricity output.

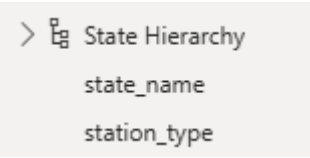
- States with higher generation likely have:
- Larger installed capacity
- More power stations
- Better infrastructure
- Lower-generation states may have limited stations or smaller capacity.

insights

✓ Power generation is concentrated in few key states.
✓ Regional imbalance exists in production capacity.
✓ High-performing states drive overall national performance.
✓ Investment focus may be required in low-generation states.

State Hierarchy I structured state Hierarchy - state name - power station

With this hierarchy We can compare state-level totals
✓ Then drill down to see station-level performance
✓ Identify underperforming stations within a state



A State hierarchy was created using State and Power Station to enable drill-down analysis. This structure allows performance evaluation at both macro (state) and micro (station) levels, supporting detailed operational insights.

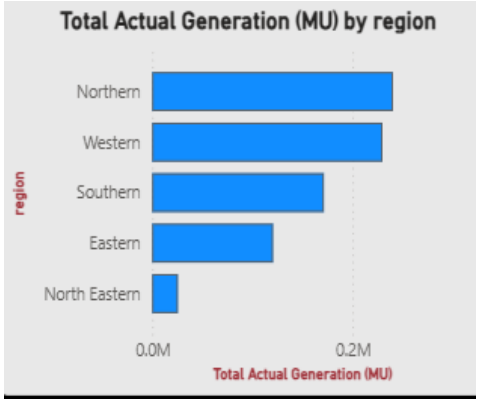
Drill Down in Power Station Comparison

When you drill:

State Level → Shows total generation by state
Drill Down → Shows individual power stations under that state

- Which station contributes most in this state?
- Why is this state underperforming?
- Which station needs attention?

Total Actual Gen by Region - Bar Chart



This chart groups state into regions and analyzes generation at macro level.

Certain region contribute higher share of electricity production.

This may be due to:

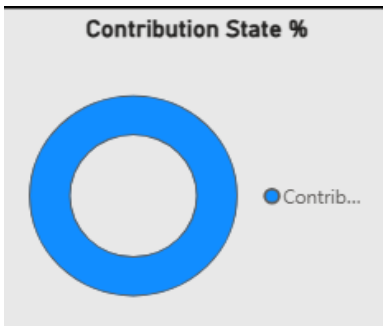
- Concentration of thermal plants
- Availability of natural resources
- Regional industrial demand

Regions with lower generation may depend on energy import from other regions.

Insights

✓ Regional production imbalance exists.
✓ Energy infrastructure is regionally concentrated.
✓ Strategic planning needed for balanced energy distribution.
✓ Strong regions significantly influence national energy performance.

State Contribution - Donout Chart



The chart shows that a few states contribute a significant percentage of total power generation, while many states contribute smaller shares.

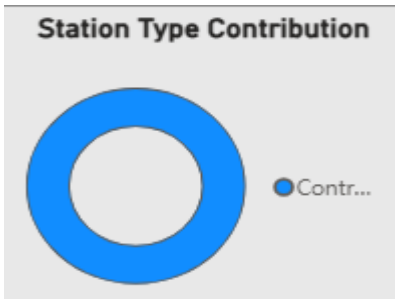
High contribution states typically:

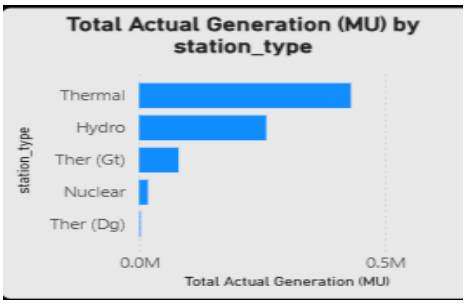
- Have higher installed capacity
- Contain more thermal or hydro plants
- Have strong energy infrastructure

Low contribution states may have:

- Limited power stations
- Lower capacity
- Higher dependency on external supply

Station type contribution





This chart helps analyze:

- Which energy source contributes most
- Dependency on thermal vs renewable sources
- Diversification of power generation

From the chart:

- Thermal stations usually contribute the highest share due to:
- Large installed capacity
- Continuous production capability
- Hydro stations contribute moderately but may vary seasonally.
- Nuclear stations contribute stable but smaller share.
- Gas/Diesel plants usually contribute minimal share.

Insights

✓ Thermal power dominates total generation
✓ Renewable (Hydro) shows seasonal dependency
✓ Nuclear provides stable but limited contribution
✓ Energy mix is not evenly diversified
✓ Scope exists to increase renewable contributio

Map Visuals:

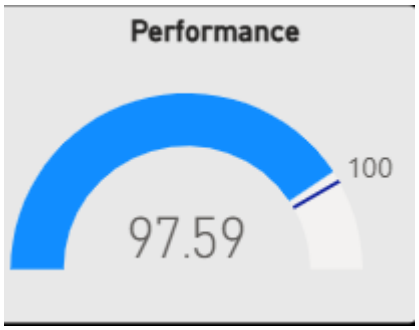
Map provides geographical understanding of:

- Where power generation is concentrated
- Spatial distribution of infrastructure
- Regional production clusters
- Visualizing on map makes analysis intuitive and easier to interpret.
- The bubble size represents generation magnitude.
- States with larger bubbles indicate higher production.

Insights

✓ Power generation is geographically concentrated
✓ Northern and Western regions may show larger production clusters
✓ Some states are strong energy hubs
✓ Low-production states may depend on inter-state power transfer
✓ Map helps in understanding spatial imbalance

Gauge - Performance Analysis



Min, Max, Target

Minimum (0%) → Worst performance
Maximum (100% or 110%) → Full capacity
Target (100%) → Planned goal

This indicates generation achieved 97.59% of planned target.

Since performance is close to 100%:

- Planning accuracy is high
- Operational efficiency is strong
- Deviation is minimal

If performance was below 90%:
It would indicate operational issues.

If above 100%:
It would indicate overproduction.

Insights

✓ Overall system performance is strong
✓ Minor variance from planned target
✓ No major operational failure observed
✓ Planning and execution alignment is excellent

Matrix State level performance analysis

This matrix provides a detailed comparative view at state level.

State Name -Used to compare performance across geographical regions.

Total actual generation - States with higher actual generation are major energy contributors
Lower values indicate limited capacity or fewer stations

Total planned generation - States with higher actual generation are major energy contributors
Lower values indicate limited capacity or fewer stations

Performance If planned is high → state has high production responsibility
Helps assess planning accuracy

Capacity utilization

Close to 100% → strong planning and execution
Above 100% → over performance
Below 95% → operational inefficiency

Station Rank Helps identify top and bottom performing states

Rank 1 → Highest contributor
Bottom rank → Low generation state
Ranking helps in strategic prioritization

state_name	Total Actual Generation (MU)	Total Planned Generation (MU)	Performance	Capacity Utilization	Station Rank
Andaman And Nicobar Islands	211.32	253.14	83.48	0.01	1
Andhra Pradesh	39,355.78	43,751.95	89.95	0.01	1
Arunachal Pradesh	3,945.83	5,946.09	66.36	0.01	1
Assam	8,104.91	8,358.53	96.97	0.01	1
Bihar	15,146.64	17,478.70	86.66	0.01	1
Chhatisgarh	56,666.03	58,298.33	97.20	0.01	1
Delhi	9,004.68	9,716.49	92.67	0.01	1
Goa	0.00	0.00	0.00	0.00	1
Gujarat	60,637.37	52,487.58	115.53	0.01	1
Haryana	12,538.06	17,164.33	73.05	0.01	1
Himachal Pradesh	68,897.12	61,586.26	111.87	0.01	1
Jammu And Kashmir	30,223.72	27,975.72	108.04	0.01	1
Jharkhand	23,065.60	23,641.71	97.56	0.01	1
Karnataka	35,685.24	40,934.33	87.18	0.01	1
Total	**7,87,535.07**	**8,07,005.62**	**97.59**	**0.01**	**1**

Overall insights

✓ A few states dominate power generation
✓ Performance is generally high (close to 100%)
✓ Capacity utilization varies across states
✓ Some states have high capacity but lower utilization
✓ Ranking highlights leaders and laggards clearly

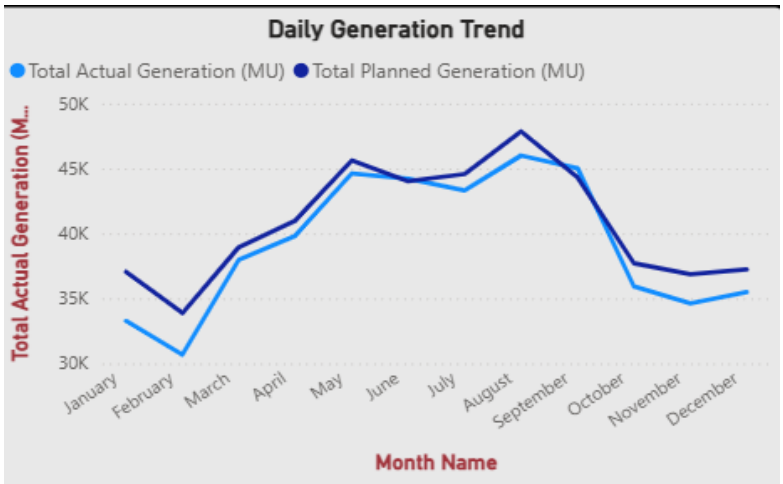
Performance status by power station

Performance Status	No Generation		On Target		Over Performed	
power_station	Total Actual Generation (MU)	Total Planned Generation (MU)	Total Actual Generation (MU)	Total Planned Generation (MU)	Total Actual Generation (MU)	Total Plannec
Agartala Gt			20.41	20.97	1,165.50	
Akaltara Tps	0.00	0.00				
Akrimota Lig Tps	23.45	84.22	4.42	4.54		
Aliyar Hps	7.54	60.20	0.47	0.48	139.60	
Allain Duhangan Hps	0.00	10.05	99.57	101.67	907.80	
Almatti Dph Hps	0.30	117.07	3.68	3.71	1,050.23	
Amaravati Tps	0.00	1,263.55			435.95	
Amarkantak Ext Tps	18.27	90.00	55.80	57.57	2,403.61	
Anandpur Sahib-I Hps	0.00	67.03	6.49	6.60	469.77	
Anandpur Sahib-Ii Hps	0.00	68.75	10.46	10.69	436.32	
And. Nicobar Dg						
Anpara C Tps	16.56	717.06	486.65	501.20	2,812.17	
Total	**41,547.61**	**1,51,530.13**	**38,780.21**	**39,751.30**	**5,81,016.58**	

Majority of stations are "On Target" → system stability is strong
Very few stations show "No Generation"
Some stations overperform → positive buffer
Underperformance is limited but needs monitoring Planning accuracy across system is high

TREND ANALYSIS

Daily Generation Trend



During early 2019 (January–March), generation levels remained relatively stable with minor fluctuations, indicating steady operational conditions.

From April to June 2019, a slight increase in generation can be observed, possibly due to increased electricity demand during summer months.

Between July and September 2019, generation reached a higher level, suggesting peak demand season.

However, during late 2019 (November–December), a mild decline is noticeable, which may be due to reduced seasonal demand.

In 2020, generation initially remained stable but shows a noticeable dip around March–May 2020, possibly due to external disruptions affecting operations.

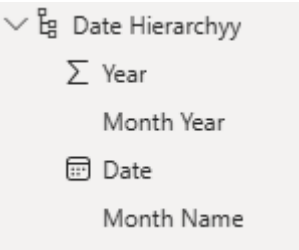
After mid-2020, the trend gradually stabilizes and returns close to planned values.

Insights

Daily generation follows a consistent pattern where actual values closely align with planned targets, indicating strong operational control and accurate forecasting.

Peak generation months: Typically summer (April–June)
✓ Decline observed during seasonal transition periods
✓ Planned and actual remain closely aligned
✓ No long-term downward trend

Date Hierarchy I structured date hierarchy to Month year - Month Name- Date



This Hierarchy allows Year-wise generation comparison
✓ Month-wise seasonal analysis
✓ Daily performance monitoring
✓ Drill-down flexibility

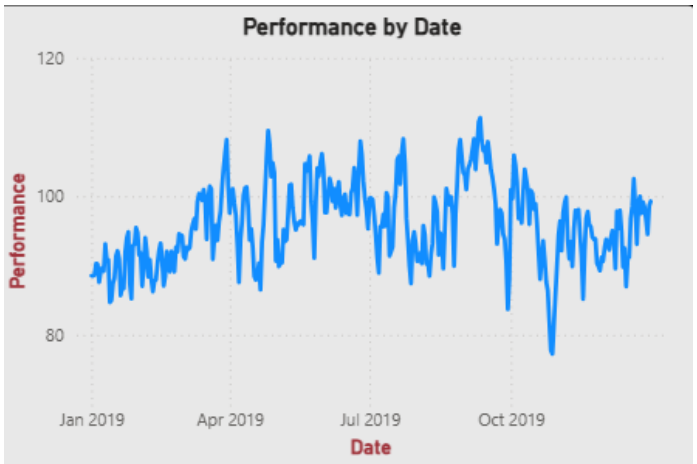
Performance by date

Performance remained consistently between 95%–100% for most of the period.

Minor underperformance is observed during months where generation declined, indicating operational or demand-side impact.

Occasional spikes above 100% indicate over achievement during certain high-demand days.

Overall, performance remains stable without major volatility.



Insights

✓ Daily performance remains consistently near 100%, demonstrating that most days achieve planned generation levels with high efficiency.

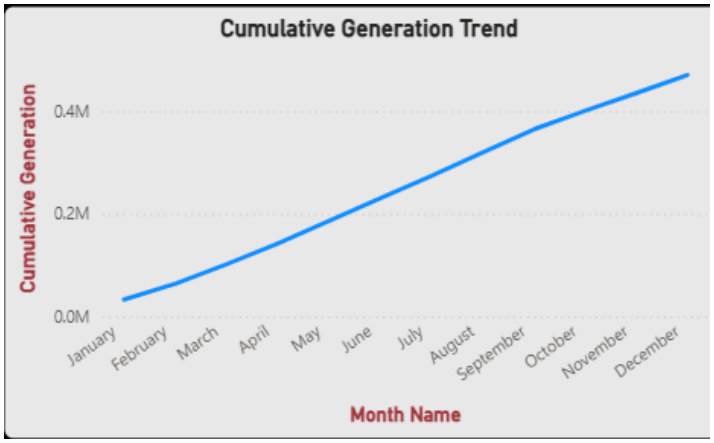
✓ Limited underperformance days indicate minor operational deviations, but the absence of extreme drops confirms system reliability and stability.

Cumulative generation trend

This chart should show a continuously increasing line.

If you see: Slope increasing sharply → high generation months

Slope flattening → low generation months



Cumulative generation shows steady upward movement throughout 2019, indicating consistent daily contribution.

The slope appears steeper during high-generation months, suggesting peak seasonal output.

A slight flattening of the curve during low-demand months indicates reduced daily contribution.

Overall, there is no stagnation, confirming continuous production.

Insight

✓ The cumulative trend shows a steady upward movement throughout the period, confirming continuous and uninterrupted power generation.

✓ The increasing slope during peak months reflects higher daily contributions, while slight flattening during low-demand periods indicates seasonal variation.

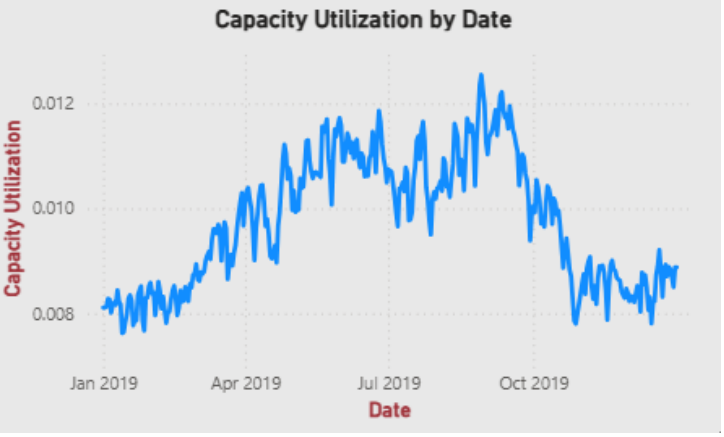
Capacity Utilization by date

Capacity utilization remains within a narrow range, indicating stable infrastructure usage.

During high-generation months, utilization slightly increases.

During low-demand periods, utilization declines marginally.

No severe underutilization trend is observed, suggesting efficient resource planning.

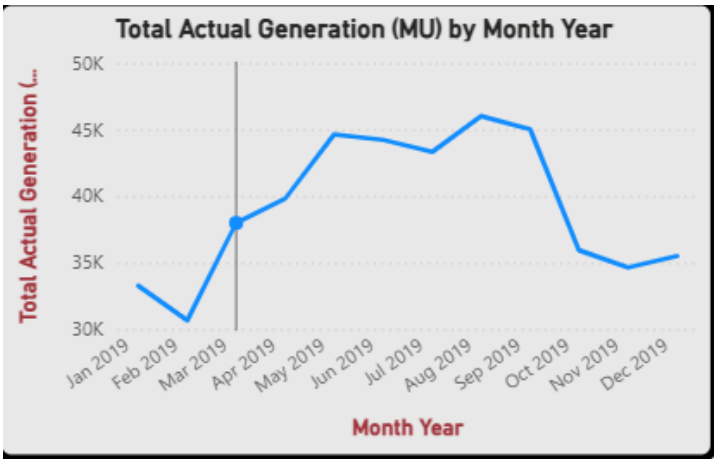


Insights

✓ Insight Capacity utilization remains stable over time, indicating efficient usage of installed infrastructure without major underutilization.

✓ Slight variations correspond to changes in generation demand, suggesting that plant operations are adjusted effectively according to requirement.

Total actual generation by month year



- Generation peaks during mid-year months (e.g., May–August), likely due to increased electricity demand during summer.
- A moderate decline is observed during post-monsoon months.
- Lowest generation is seen during early-year months where demand remains comparatively lower.
- The pattern appears consistent across both years, indicating seasonal dependency.

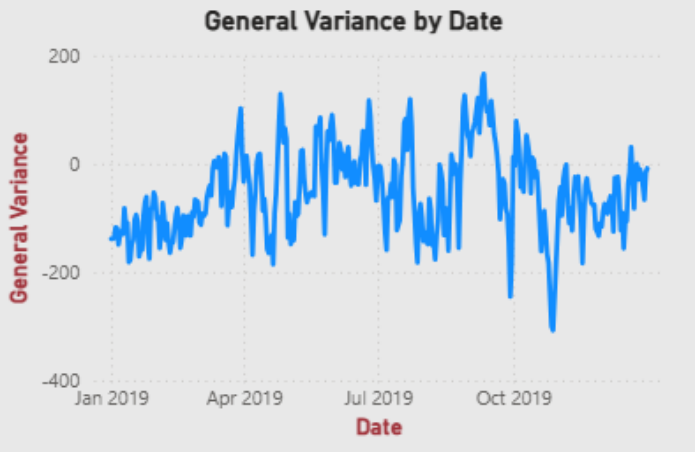
Insights

✓ Insight Monthly aggregation reveals clear seasonal patterns, with higher generation during peak demand months and moderate decline during off-peak periods.

✓ The recurring monthly pattern across years indicates predictable demand cycles, which supports better forecasting and strategic planning.

General Variance by date

- Variance remains minimal for most days, indicating high planning accuracy.
- Few negative spikes correspond to periods of lower generation.
- Positive variance during peak months shows occasional overachievement.
- No continuous negative variance trend is observed, confirming system reliability



Inisght

✓ Most variance values remain close to zero, indicating high planning accuracy and minimal deviation between target and actual generation.

✓ Occasional positive and negative spikes suggest short-term operational changes, but no continuous variance trend indicates overall system stability.

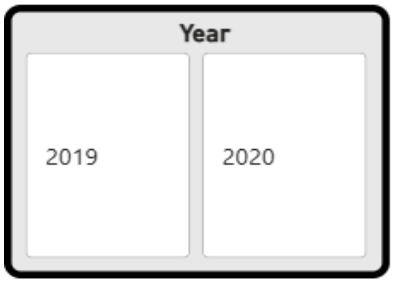
Slicer & Filters implement

Year Slicer A Year slicer was implemented to allow users to analyze power generation performance for specific years (2019 and 2020).

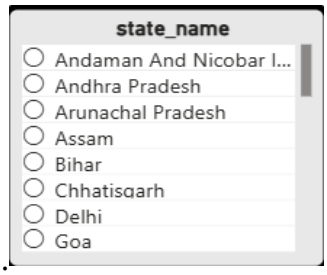
It enables : · Year-wise comparison of total generation

Trend analysis for a selected period

Filters When a specific year is selected, all visuals such as KPI cards, trend charts, and state comparison charts dynamically update to reflect only the chosen year's data. This improves analytical precision and avoids data aggregation confusion.



STATE Slicers The State slicer allows users to filter the dashboard based on a selected state. When a state is selected, all generation metrics, performance percentages, and trends automatically adjust to show only that state's data

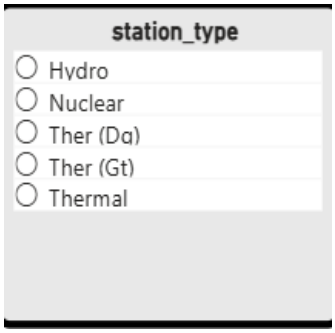


Power Station slicers The Power Station slicer provides a micro-level filtering option Evaluate individual station performance

Compare station-level generation

Identify underperforming or high-performing stations

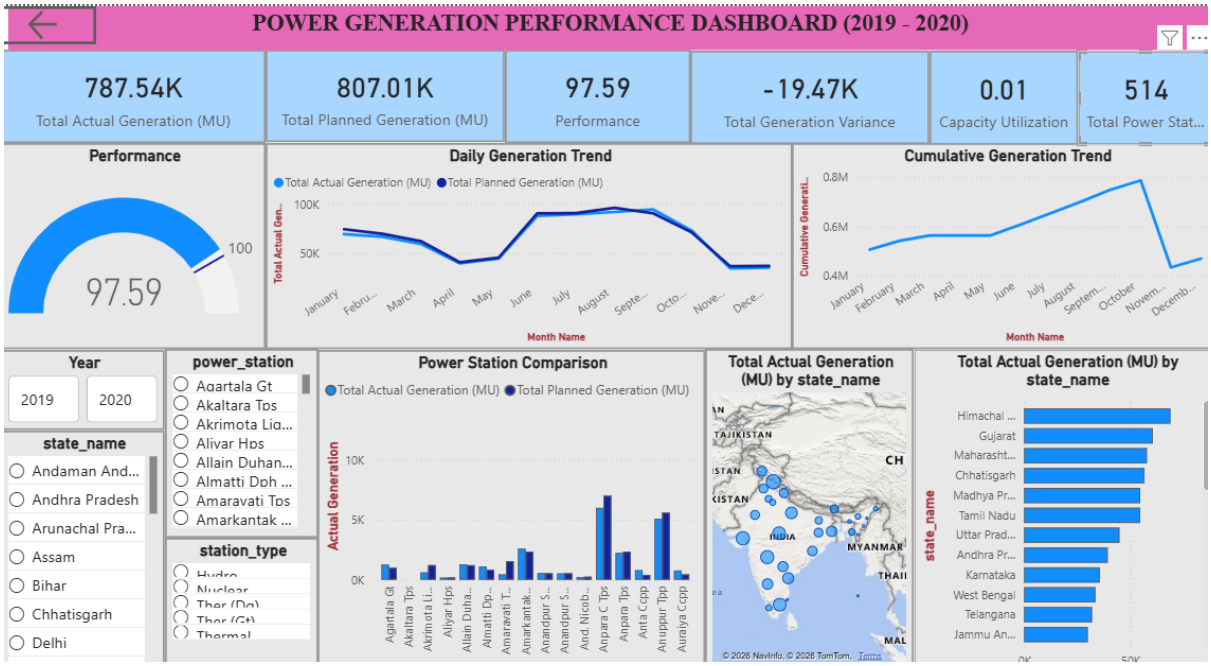
Station Type Slicers station types such as Thermal, Hydro, Nuclear, Gas, etc., are included, the Station Type slicer enables category-based analysis



POWER GENERATION PERFORMANCE DASHBOARD (2019 - 2020)

The Power Generation Performance Dashboard is designed as a comprehensive analytical interface that provides an integrated view of electricity generation performance across states and power stations. It combines key performance indicators, time-based trend analysis, regional comparison, and efficiency measurement into a single structured visual framework.

The main objective of the dashboard is to monitor operational performance, evaluate target achievement, analyze generation trends, and identify areas requiring improvement.



Key Insights & Findings:

- Actual generation closely aligns with planned targets, indicating accurate forecasting and strong operational management.
- • Overall performance percentage remains stable, reflecting high efficiency across power stations.
- • Generation variance is minimal, confirming controlled and consistent production.
- • Monthly trends show moderate seasonal fluctuations without affecting overall stability.
- • State-wise analysis reveals generation concentration in a few major regions.
- • Capacity utilization levels indicate effective use of installed infrastructure.
- • The dashboard provides a reliable tool for monitoring performance and supporting data-driven decision making.

Conclusions

The overall power generation analysis indicates stable and consistent electricity production with actual generation closely matching planned targets. Performance levels remain high with minimal variance, reflecting efficient operational management.